Ji Won **Chung**

Researcher · Long-term Computing · Visual Systems

g1isgone.github.io | ji_won_chung@brown.edu | jiwonchung

I research and design systems for long-term routine management and sensemaking.

Education

Brown University - Ph.D. Candidate, Computer Science — *Providence, RI*

Cumulative GPA: 4.00/4.00 - Advisor Jeff Huang — *Sep 2021 - May 2027*

Related Coursework: Advanced Graphics, Computer Vision, Deep Learning, UI/UX, HCI Seminar, Sheridan Teaching Seminar

Research & Work Experience

Brown Human Computer Interaction Lab — *Providence, RI*

PhD Candidate - Advisor Jeff Huang — *Sep 2021 - Present*

- Developing productivity tools and new scheduling algorithms that account for both one's biological, hormonal cycle (menstruation) and external constraints (vocation) **$20,000 Funding**, First Place in Brown-Lifespan Center for Digital Health's Pitch Competition
- Implemented a daily sleep regularity model and analyzed routines on 10 years of data with FFT for 1 Million active users of Sleep as Android
- Developed new, visual analytic tools to gain insights on personal day-to-day routines from personal messaging data for Sochiatrist
- Extracted sleep and heart data from sleep tracking devices (i.e. Whoop Band, Withings Mat) with mitmproxy and apktool to investigate data validity and evaluate real-world feasibility of in-situ tracking for elderly patients with Dementia, in collaboration with IMPACT Collaboratory
- Enabling new story-telling paradigms with 2.5D anamorphic and parallax interactions on the web via JavaScript and iPad
- Investigation of collaborative visual systems and user dynamics with augmented reality filters with Unity, Meta Quest Pro, and Spectacles

Apple — *Cupertino, CA*

Human Factors Research Intern - Robert Deloatch | Human Engineering — *May 2025 - Sep 2025*

- Conducted Human Factors Evaluation to provide design feedback that improve user experience and retention for Ambient Music feature
- Designed and conducted an internal 8-person user study and Human Factors expert evaluation to identify usability issues with NFC Tap to Add Card
- Designed a 25-person user study with semi-structured interviews and behavioral observations to evaluate user experience on Adaptive Power, an iOS 26 feature that optimizes battery life through behavioral learning and performance adjustments in collaboration with the Core OS team
- Prototyped an internal data annotation and interactive visualization dashboard powered by LLMs to understand AI tool integration with internal team and improve internal Human Factors researcher workflow with rapid, iterations of visualizations and data analyses using Flask and React

Adobe Research — *San Jose, CA*

Research Scientist Intern - Ryan Rossi | Core Technologies, Data Science Lab — *May 2023 - Aug 2023*

- Developed a body-driven, data-performance system with iPhone, Swift, and ARKit; evaluated by 30 public participants

Morgan Stanley — *New York, NY*

Software Engineer - Institutional, Listed Sales & Trading — *Aug 2018 - May 2021*

- Developed new swap-hedge aggregation strategies, routing features, regression and software release mangers for client migration and tested load capacity for real-time electronic sales and trading systems for high-volume, institutional trades in C++, Python, Jasmine, Angular, and XML

Graph-based Matching for Word Spotting in Handwritten Documents — *Northampton, MA*

Senior Thesis - Nick Howe | Awarded Highest Honors — *Sep 2017 - May 2018*

- Proposed a new bidirectional, one-shot matching algorithm to increase query retrieval accuracy for handwritten words with MATLAB

Human Computation & Visualization Laboratory — *Northampton, MA*

Research Assistant - R. Jordan Crouser — *May 2016 – May 2017*

- *DSMVis: Interactive Visual Exploration of the DSM-5 for Mental Health Providers*, an interactive diagnoses filtering system via bubble charts to reduce diagnostic bias of mental health clinicians and organizational bias of the DSM-5 using D3.js, HTML, and CSS Honorable Mention Best Poster

Pacific Northwest National Laboratory — *Seattle, WA*

National Security Internship Program — *Jun 2017 - Aug 2017*

- Developed an interactive, web-based visualization tool to discover anomalies and patterns in graphs motifs in React and D3.js

Publications

J. Chung, R. Yuan, K. Zitting, J. Chen, N. Xu, N. Daskalova, J. Huang. 2025. "Beyond the Circadian Rhythm: Variable Cycles of Regularity Found in Long-Term Sleep Tracking". In CHI Conference on Human Factors in Computing Systems (CHI '25), April 26-May 1, 2025, Yokohama, Japan. ACM, New York, NY, USA, 18 pages. doi: 10.11453706598.3713868

J. Chung, T. Zhou, I. Chen, K. Hsu, R. A. Rossi, A. Siu, S. Guo, F. Dernoncourt, J. Tompkin, J. Huang. 2025. "InfoVids: Reimagining the Viewer Experience with Alternative Visualization-Presenter Relationships". arXiv: 2505.03164

T. Zhou, J. Yang, V. H. Chan, **J. Chung**, J. Huang. 2024. "PortalInk: 2.5D Visual Storytelling with SVG Parallax and Waypoint Transitions". User Interface Software and Technology (UIST '24), October 13–16, 2024, Pittsburgh, PA, USA. ACM, New York, NY, USA, 16 pages. doi:10.1145/ 3654777.3676376

J. Chung, J. Fu, Z. Deocadiz-Smith, M. F Jung, and J. Huang. 2023. "Negotiating Dyadic Interactions through the Lens of Augmented Reality Glasses". In Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23). Association for Computing Machinery, New York, NY, USA, 493–508, doi: 10.1145/3563657.3595967

N. Howe and **J. Chung**, "Symmetric Inkball Alignment with Loopy Models," *2019 International Conference on Document Analysis and Recognition (ICDAR)*, Sydney, Australia, 2019, pp. 349-354, doi: 10.1109/ICDAR.2019.00063

Awards & Honors

First Place - $20,000	Digital Health Pitch Competition - Productivity, Period	*Sep 2024*
Finalist	Digital Health Pitch Competition - Sleep Regularity in the Wild	*Jun 2022*
Global Excellence Award - Giving Back	Awarded by Morgan Stanley for Volunteering on Billion Oyster Project, Deployed UI App for Oyster Data Collection	*Oct 2022*
Honorable Mention Best Poster	IEEE Visual Analytics Science and Technology Conference, DSMVis: Interactive Visual Exploration of the DSM-5 for Mental Health Providers	*Aug 2017*

Teaching Experience

Brown University — *Providence, RI*

Graduate Teaching Assistant - Interaction Design — *Jan 2025 - May 2025*

- Designed assignment and studio grading rubrics and helped assignment development for 110 person class

Smith College — *Northampton, MA*

Teaching Assistant - Data Structures & Microprocessors and Assembly — *Sep 2017 - Dec 2017*

- Assisted students resolve conceptual and programming questions during studio and office hours

Research Mentoring

Audrey Chou	Undergraduate, Brown University	*Jan 2025 - Present*
Emily Lin	Undergraduate, Brown University	*Aug 2024 - Present*
Rebecca Chou	Undergraduate, Brown University	*Jan 2025 - Sep 2025*
Ivery Chen	Undergraduate, Brown University & RISD - Software Engineer at Tesla	*Feb 2023 - May 2024*
Kevin Hsu	Undergraduate, Brown University - Software Engineer at Stripe	*May 2022 - May 2024*
Neil Xu	Undergraduate, Brown University	*Feb 2022 - May 2022*
Lucas Weissman	Undergraduate, Williams College, exploreCSR Program	*Feb 2022 - May 2022*
Jiahua Chen	Undergraduate, Brown University	*Sep 2021 - Dec 2022*

Service

Reviewing

CHI 2026	Paper Reviewer (x3)	*Oct 2025*
DIS 2025	Paper Reviewer (x1) — Special Recognition for Outstanding Review	*Mar 2025*
CHI 2025	Paper Reviewer (x5) — Special Recognition for Outstanding Review	*Oct 2024*
CHI 2024	Paper Reviewer (x3)	*Oct 2023*
IMWUT 2024	Paper Reviewer (x2)	*Feb 2024*
IEEE VR 2024	Paper Reviewer (x1)	*Nov 2023*
CHI 2023	Helped Paper Review (x1)	*Oct 2022*
SIGGRAPH 2022	Poster Reviewer (x2)	*May 2022*
UIST 2022	Paper Reviewer (x1)	*May 2022*
CHI 2022	Helped Paper Review (x1)	*Oct 2021*

Brown University

Ph.D. Admissions	Student Application Reader for 90 Ph.D. student applications	*Jan 2024*
GWICS+	Lead Organizer of Graduate Women in CS+	*Feb 2022 - May 2024*
D&I Committee	PhD Student Representative of the Diversity and Inclusion Committee	*Sep 2021 - May 2024*
exploreCSR	PhD Mentor for undergraduates pursuing Socially-Responsible Artificial Intelligence	*Feb 2022 - May 2022*

Posters

J. Chung, I. Raut, J. Y. Yun, K. Pien, S. Sridhar, M. R. Crouser, and R. J. Crouser, "DSMVis: Interactive visual exploration of the DSM5 for mental health providers," *2017 IEEE Conference on Visual Analytics Science and Technology (VAST)*, Phoenix, AZ, USA, 2017. **Honorable Mention Best Poster**.

J. Chung, "Dynamic Network Analysis via Motifs (DYNAMO) Software Development," *2017 Pacific Northwest National Laboratory National Security Internship Program Research Symposium*, Richland, WA, USA, 2017. Presentation.

J. Chung, Z. Rizvi, S. Sridhar, and J. Y. Yun, "A Business Opportunity: Targeting Expedia's Niche Market in Travel Packages Via Analytical and Predictive Modeling," *2017 Electronic Undergraduate Statistics Research Conference (eUSR)*, 2017. **Third Placed Paper in USCLAP Competition in Intermediate Statistics.** Presentation.